

1. Administrative & Study Identification

Full Study Title: A Study of Coformulated Favezelimab/Pembrolizumab (MK-4280A) Versus Physician's Choice Chemotherapy in PD-(L)1-refractory, Relapsed or Refractory Classical Hodgkin Lymphoma (MK-4280A-008) **Short Title/Acronym:** KEYFORM-008 **Protocol Number:** MK-4280A-008 **NCT Number:** NCT05508867 **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Coformulated Favezelimab/Pembrolizumab (MK-4280A) **Phase of Development:** Phase 2 **Trial Status:** Active, Not Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of MK-4280A compared to physician's choice of chemotherapy in terms of Progression-Free Survival (PFS) per Lugano Response Criteria as assessed by investigator. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DOR), and the safety and tolerability of the study interventions.

2.2 Background & Rationale Researchers are looking for a way to treat classical Hodgkin lymphoma (cHL) that is relapsed (the cancer has come back after treatment) or refractory (current treatment has stopped working to slow or stop cancer growth). The importance of PD-1 therapy in relapsed or refractory (R/R) classical Hodgkin Lymphoma is well established. However, most patients eventually develop progressive disease, and optimal therapy after anti-PD-1 failure remains a significant unmet medical need. Researchers want to learn if people who receive coformulated favezelimab/pembrolizumab (MK-4280A) live longer without the cancer getting worse compared to those who receive chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Physician's choice chemotherapy: bendamustine or gemcitabine) **Blinding:** Open-label **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Target N\$: 203 participants **Number of Sites:** Global Multicenter **Estimated Study Start:** 18-Oct-2022 **Estimated Primary Completion:** 03-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histologically confirmed diagnosis of classical Hodgkin lymphoma (cHL) that is 2-fluorodeoxyglucose-avid (FDG-avid). 2. Has relapsed (defined as disease progression after most recent therapy) or refractory (defined as failed to achieve CR or PR to most recent therapy) cHL and exhausted all available treatment options with known clinical benefit. 3. Has progressed on treatment with an anti-PD-(L)1 monoclonal antibody (mAb) administered either as monotherapy or in combination with other checkpoint inhibitors or other therapies. 4. Submits an archival (≤ 5 years) or newly obtained tumor tissue sample which has not been previously irradiated.

4.2 Exclusion Criteria 1. Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy or any other form of immunosuppressive therapy. 2. History of central nervous system (CNS) metastases or active CNS involvement. 3. Has an active autoimmune disease that has required systemic treatment in past 2 years except replacement therapy. 4. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 5. Has an active infection requiring systemic treatment. 6. History of hemophagocytic lymphohistiocytosis. 7. Has an active seizure disorder that is not well controlled. 8. Has clinically significant (ie, active) cardiovascular disease. 9. Received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 10. Received prior radiotherapy within 2 weeks of start of study intervention or radiation related toxicities requiring corticosteroids. 11. Has not adequately recovered from major surgical procedure. 12. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 13. History of human immunodeficiency virus (HIV). 14. Has had an allogeneic hematopoietic stem cell or solid organ transplantation within the last 5 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-4280A arm: 800 mg favezelimab and 200 mg pembrolizumab administered on Day 1 of each cycle. Control arm: Physician's choice of bendamustine (90-120 mg/m²) or gemcitabine (Trade names: Gemzar, Infugem; 800-1200 mg/m²). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Once every 3 weeks (Q3W) for up to 35 cycles (approximately 2 years) for MK-4280A, or up to 6 cycles for the chemotherapy control arm.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms and treatment-related toxicities. **Prohibited:** Concurrent investigational

agents, chronic systemic corticosteroids, and other systemic anti-cancer therapies during the active study phase.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Tissue Sample Submission
Baseline	Day 0	Randomization, Baseline Assessments, First Dose
Interim	Q3W (Cycle 1-35)	Safety Labs, IV Infusion, PK/PD Sampling, Tumor Imaging (every 12 weeks)
End of Study	30 Days Post-Last Dose	Final Vital Signs, Exit Interview, Resolution of AEs, Transition to Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS). PFS is defined as the time from randomization to the first documented disease progression per Lugano criteria 2014 or death due to any cause, whichever occurs first. **Measurement Method:** Assessed per Lugano Response Criteria by investigator.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Objective Response Rate (ORR) per Lugano Response Criteria as Assessed by investigator * Duration of Response (DOR) per Lugano Response Criteria as Assessed by investigator * Safety and Tolerability (incidence of Adverse Events per NCI CTCAE v5.0)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any untoward medical occurrences graded per NCI CTCAE v5.0 criteria. **Serious Adverse Events (SAEs):** Evaluated and reported to the sponsor and regulatory authorities within standard safety reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) to evaluate safety and efficacy signals throughout the study execution.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for efficacy endpoints; Safety Set (all treated participants) for safety endpoints. **Sample Size Justification:** The targeted enrollment is 203 patients. **Handling of Missing Data:** PFS, OS, and DOR estimated using the Kaplan-Meier method. ORR with 95% CI estimated using the Clopper-Pearson method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine remote and onsite monitoring visits to verify protocol compliance, source data, and investigational product accountability. **Audit Trail:** Secure Electronic Case Report Form (eCRF) utilizing comprehensive audit trails, data clarification forms, and centralized data management systems.

11. Study Identifiers & Governance

Primary ID: MK-4280A-008 **Posting ID:** 273647628979273717 **Registry Number:** NCT05508867 **Sponsor/Lead Organization:** Merck **Study Title:** KEYFORM-008 **Official Regulatory Title:** A Phase 2 Randomized Clinical Study of MK-4280A (Coformulated Favezelimab [MK-4280] Plus Pembrolizumab [MK-3475]) Versus Physician's Choice Chemotherapy in PD-(L)1-refractory, Relapsed or Refractory Classical Hodgkin Lymphoma (KEYFORM-008)

12. Clinical Framework (Classical Hodgkin Lymphoma Specific)

Primary Condition: Relapsed or Refractory Classical Hodgkin Lymphoma / Hodgkin Disease **Diagnosis Threshold:** Histologically confirmed FDG-avid R/R cHL **Medical Methodology:** Dual Immune Checkpoint Blockade (LAG-3 and PD-1 inhibitors) vs Chemotherapy **Optimization Target:** Restoration of T-cell effector function and prolonged tumor response

13. Study Procedures & Methodology

Management Pathway: Monoclonal antibody systemic administration **Education Protocol (HCP):** Clinical trial protocols for irAE (immune-related adverse events) management **Education Protocol (Patient):** Immunotherapy side-effect awareness **Quality Audit Mechanism:** Assessment of radiographic imaging for objective tumor response per Lugano criteria

14. Observation & Follow-Up Schedule

Total Duration: Up to ~2 years (35 cycles of 3 weeks) + subsequent survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Day 1 of each treatment cycle) **Remote Monitoring Frequency:** Routine telemedicine and outcome tracking as dictated by the site **Checklist Review Interval:** Tumor imaging assessment performed approximately every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					